OntoGeoRAG: Ontology-Constrained Knowledge Graph Construction from Geological Literature via Retrieval-Augmented Generation

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Abstract

Geological knowledge graphs link geological objects, processes, and observations into structured, queryable representations, but constructing them at scale is difficult: geological concepts are expressed heterogeneously across the scientific literature and require expert interpretation to organize into formal relations. Existing graphs are built manually, which limits their scope and makes them costly to update as new literature appears. We present OntoGeoRAG, a pipeline that automatically constructs a geological knowledge graph from scientific literature by combining ontology-guided query generation, text retrieval, language-model extraction, and evidence-based verification. The approach requires no domain-specific model training and runs on openly available language models. The pipeline is evaluated on 41 peer-reviewed papers on mass-transport deposits, large-scale gravity-driven sediment failures common on continental margins, using an expert-curated ontology as both the extraction schema and the evaluation benchmark. Without retrieval, the language model recovers 26% of benchmark relations from memory alone, establishing a parametric baseline. Adding keyword-based retrieval raises recall to 47%. Replacing keyword retrieval with a semantically-aware re-ranking step increases recall further to 77%, recovering eight relation types that keyword retrieval had consistently missed. The final graph contains 153 relations, of which the 101 highest-confidence triples contain no unsupported extractions under an automated verification protocol. An independent verification experiment using a second language model confirms this result, with 96% agreement between the two models. The central finding is that the primary performance bottleneck is retrieval quality, not the language model's reasoning ability. These results suggest that future improvements should prioritize retrieval rather than model scaling, and provide a reproducible approach to literature-grounded geological knowledge extraction, with potential for extension to other geological domains where a comparable expert ontology exists.

1 Introduction

1.1 Structured Knowledge as a Bottleneck in Geoscience Interpretation

Geological interpretation of subsurface data is fundamentally a knowledge-driven task. Whether identifying sedimentary bodies on seismic profiles, correlating stratigraphic units across a basin, or assessing geohazard potential from slope morphology, a geoscientist draws on a large body of accumulated knowledge, which observable features correspond to which geological objects, which processes produced them, and which environmental conditions govern their occurrence. This knowledge exists, it has been documented across decades of peer-reviewed literature but it is distributed, heterogeneous in terminology, and not organized into any structure that can be queried, updated, or integrated computationally.

This fragmentation is a practical obstacle. In seismic interpretation, the vocabulary linking observable reflection patterns to depositional processes differs between research groups, basins, and decades of publication. In stratigraphy, the relationships between depositional environments, lithological signatures, and sequence boundaries are encoded inconsistently across regional case studies. In geohazard assessment, the links between precursor signals and slope instability have been documented piecemeal across observatory reports and journal articles that were never designed to be compared systematically. In each case, a geoscientist wishing to draw on the full body of available knowledge must perform their own literature synthesis a process that is slow, non-reproducible, and impossible to scale as the literature grows.

Knowledge graphs address this problem directly. A knowledge graph encodes domain facts as a network of entities connected by labeled relations, enabling structured queries, automated reasoning, and cross-domain integration that unstructured text cannot support. Their potential for geoscience has been demonstrated across several domains. In mineral exploration, knowledge graphs linking geological units, geochemical anomalies, and deposit types have enabled spatial reasoning over exploration targets at scales that manual comparison cannot reach (**?**). In global stratigraphic and paleontological research, the Deep-time Digital Earth initiative is assembling a cross-domain geoscience knowledge graph that integrates datasets from international databases to support queries spanning geological time and geographic regions (**?**). Reviews of knowledge graph applications across the geosciences document further use cases in mineralogy, petrology, and remote sensing, and consistently identify manual curation as the dominant construction method and the central scalability bottleneck (**?**).

The cost of manual curation is substantial. Constructing a knowledge graph from even a small literature corpus requires a domain expert to read each paper, identify relevant entities and relations, reconcile conflicting terminology across sources, and encode the result in a formal structure. This typically takes months and produces a static snapshot that becomes outdated as new literature appears. Extending the graph to new geological objects, new basins, or new interpretation questions requires repeating the process from scratch. Automating this construction, even partially, would remove a significant barrier to deploying structured geological knowledge at scale.

1.2 Language Models and the Possibility of Automated Knowledge Extraction

Recent advances in large language models (LLMs), software systems trained on large text corpora that can parse, summarize, and generate natural language have opened new possibilities for automated knowledge extraction from scientific literature. LLMs can identify entities, recognize semantic relationships between them, and produce structured output from unstructured text without requiring hand-coded rules or task-specific labeled training data. Several research groups have demonstrated that this capability can be directed toward knowledge graph construction in general scientific domains (**??**), and domain-adapted models such as K2 (**?**) show that pre-training on geoscience corpora can further improve performance on disciplinary tasks.

However, two structural difficulties limit straightforward application of LLMs to geological knowledge extraction.

The first is *schema control*. Without explicit constraints, LLMs produce heterogeneous and often redundant output: the same geological relationship may be expressed through twenty different verb phrases, or collapsed into a generic category such as *related to* that carries no interpretable information. In our own preliminary experiments, between 29% and 55% of relations extracted without schema constraints fell into this uninformative category. Guiding extraction with an explicit ontological schema is necessary to ensure that extracted relations are meaningful and comparable.

The second difficulty is *retrieval grounding*. An LLM cannot process an entire literature corpus in a single operation: it must be provided with the specific text passage most likely to contain the target information. If the retrieval component the system that selects which passages to present to the model fails to surface the right passage, the relation it contains will never be extracted, regardless of the model's reasoning capability. This failure mode is particularly acute in geological literature, where the same seismic characteristic may be described as *high-amplitude*, *strong reflector*, or *acoustically bright* depending on the author, the basin, and the decade. Keyword-based retrieval methods, which match query words to passage words, cannot bridge this vocabulary gap. As this paper demonstrates quantitatively, retrieval failure not language model reasoning is the dominant performance bottleneck in automated geological knowledge extraction. Several approaches have been proposed to address these difficulties in general scientific domains. One class of methods extracts relations freely and then checks whether they are supported by the source text using a secondary language model (**?**); this improves reliability but does not constrain extraction to a predefined relation vocabulary. A second class constrains extraction from the start by including the target schema directly in the model's instructions, without any additional training (**??**); schema constraints alone have been shown to substantially reduce the proportion of uninformative or redundant extracted relations. The most recent systems combine ontology constraints with a retrieval step that selects relevant passages before extraction begins (**??**), and demonstrate that this combination outperforms either approach alone.

Our pipeline follows this last direction. The key difference is that we apply it to a geological domain with

a well-defined expert ontology, evaluate it against a manually curated benchmark, and specifically analyze what happens when retrieval fails a question that none of the above systems have examined in detail. Geoscience-specific language models, such as K2 (**?**), demonstrate that domain pre-training improves performance on geoscientific tasks, but require extensive domain-specific training data and infrastructure that most research groups cannot access. OntoGeoRAG is designed to achieve useful performance using openly available general-purpose models, without any domain-specific fine-tuning.

1.3 Mass-Transport Deposits as a Test Case with a Quantitative Benchmark

We evaluate our approach on mass-transport deposits (MTDs): large-scale sediment failures in which material moves downslope under gravity on continental margins. MTDs are a practically important target. They occur across a wide range of settings, from shallow-water canyon-head failures on the Amazon Fan and Mediterranean margins to deep-water slope systems in the North Atlantic, spanning water depths from ∼500 to ∼3500 m and thicknesses from tens to hundreds of meters (**??**). Their identification and characterization has direct relevance to seismic hazard assessment, hydrocarbon reservoir evaluation, and submarine infrastructure planning.

Identifying MTDs on conventional marine seismic reflection profiles (2D/3D; bandwidth ∼25–100 Hz; vertical resolution ∼10–40 m) relies on a precise vocabulary of seismic descriptors *chaotic*, *transparent*, *blocky*, *high-amplitude* used to characterize their internal architecture and depositional context. These descriptors and their causal relationships are documented across a substantial body of literature, but have not been organized into a unified, queryable knowledge structure.

A carefully constructed exception exists: **?** synthesized 41 peer-reviewed publications into a knowledge graph linking MTD seismic descriptors to the geological processes that produce them, yielding 88 nodes and 173 directed edges across seven MTD property categories. This graph required several months of specialist effort and represents, to our knowledge, the most complete structured encoding of MTD seismic interpretation knowledge available. We use it here not as a starting point for our pipeline, but as a ground-truth benchmark: since it was constructed manually from the same 41 papers our pipeline will process, it provides a direct and quantitative measure of how much of an expert-curated geological knowledge graph can be recovered automatically from the same literature.

1.4 Contributions of This Work

This paper introduces OntoGeoRAG, an ontology-constrained retrieval-augmented generation pipeline for automated geological knowledge graph construction. The pipeline combines four components: ontology-guided query generation, which derives targeted retrieval queries directly from the expert schema; two-stage text retrieval, which pairs keyword-based search with cross-encoder re-ranking to locate semantically relevant passages despite vocabulary mismatch; language-model triple extraction, which uses an openly available 7-billion-parameter model to convert retrieved passages into structured relation statements under schema constraints; and evidence-based verification, which checks each extracted triple against its source passage and separates well-grounded extractions from unsupported ones.

The pipeline requires no domain-specific model training, runs on a single GPU in approximately 20 minutes, and is fully reproducible from openly available models and code.

The central empirical contribution is a systematic analysis of where the pipeline succeeds and where it fails. The experiments show that retrieval quality, not language model reasoning, is the primary factor limiting relation recovery in this setting. Replacing keyword-only retrieval with semantic re-ranking raises recall from 50% to 69% against the benchmark, recovering eight relation types that keyword retrieval had consistently missed. A complementary experiment confirms that the model recovers only 35% of benchmark relations from parametric memory alone, establishing that retrieval grounding is a necessary rather than incidental component of the pipeline.

These findings have implications beyond the MTD domain: within this experimental setting, retrieval quality yielded larger gains than any change to the generative model; whether this finding generalizes to other geological domains or larger corpora remains an open question.

2 Method

OntoGeoRAG constructs a geological knowledge graph from scientific literature through five sequential steps: targeted query generation from an expert ontology, passage retrieval from an indexed corpus,language-model extraction of structured relation triples, evidence-based verification of each triple against its source passage, and assembly of verified triples into a tiered graph. Each step addresses a specific failure mode identified during pipeline development. Without ontology-guided queries, a language model produces heterogeneous and often uninterpretable relations. Without retrieval grounding, it cannot access the specific passages where relevant geological relations are stated. Without verification, it may generate plausible but unsupported statements drawn from parametric memory rather than from the corpus. Without post-processing, the same relation may appear under multiple surface forms and inflate the apparent graph size.The overall workflow is illustrated in Figure 1. Model names, hardware, and parameter choices are reported in Section 3.

results/fig01_pipeline_overview.png

Figure 1: Architecture of the OntoGeoRAG pipeline. The 41-paper corpus is segmented into overlapping text chunks and indexed for BM25 keyword retrieval. Ontology-guided queries derived from the **?** schema (88 nodes, 173 edges) are used to retrieve relevant passages under two configurations: **LLM-BM25** passes the five highest BM25-scoring chunks directly to the extractor; **LLM-Rerank** first retrieves twenty candidates and re-scores them with a cross-encoder before selecting the top five. A language model (Qwen 2.5-7B-Instruct) extracts candidate relation triples from the retrieved passages in two passes (deterministic pass A and low-temperature pass B). Each triple is verified against its source passage and assigned a support verdict (strong, weak, or not supported). Verified triples are normalized and assembled into a two-tier knowledge graph: Tier-1 contains triples confirmed in both extraction passes (101 triples); Tier-2 contains triples confirmed in one pass only (153 triples total). A stratified sample of Tier-1 triples is evaluated by a domain geoscientist (expert validation box). The effect of the LLM-BM25-to-LLM-Rerank re-ranking step on recall is reported in Section 4.2.

2.1 The Ontology as Extraction Schema and Evaluation Target

The pipeline is built around the geological ontology of **?**, referred to here as LB2019. This ontology was constructed by a team of geoscientists who read 41 peer-reviewed papers on mass-transport deposits and encoded the relationships they found into a formal graph, yielding 88 nodes representing geological objects, processes, and seismic descriptors, connected by 173 directed edges across seven categories of MTD properties: global environment, morphology, position, basal surface, upper surface, internal facies distribution, and headscarp.

We use this ontology in two ways simultaneously. First, it defines *what* the pipeline is allowed to extract: the entity and relation types in the graph become the schema that constrains the language model's output, preventing it from producing free-form or domain-irrelevant relations. Second, it provides the *evaluation benchmark*: since LB2019 was built manually from the same 41 papers, we can directly compare what the pipeline recovers to what a domain expert found, without any circularity between training and evaluation.

To make the ontology usable as an extraction schema, we reduce it to five entity types `SeismicObject`, `Process`, `Descriptor`, `Setting`, and `Evidence` and eleven relation types (Table 1). This reduction is necessary because the full ontology contains overlapping and abstract relations that span multiple sentences or require cross-paper synthesis, and cannot be extracted from individual passages. The eleven relation types cover the main causal, descriptive, and structural links in the original ontology.

For each relation type, Table 1 specifies whether both argument entities must belong to a predefined geological vocabulary (*closed-world*), or whether at least one argument may be a term used consistently in the literature but not anticipated in the original ontology (*semi-open*). The `hasDescriptor` relation is closed-world because the set of recognized seismic descriptors is finite and explicitly defined in LB2019. The `occursIn` relation is semi-open because new depositional settings may be named in papers published after the ontology was constructed.

The geological vocabulary used for validation contains more than 600 curated terms covering recognized geological objects, processes, and descriptors. It was assembled by extracting entity names from LB2019 and expanding them with common synonyms and abbreviations found in the 41-paper corpus.

The evaluation benchmark consists of 34 reference edges selected from LB2019 by applying three criteria: both argument entities must appear in the geological vocabulary; the relation type must map to one of the eleven allowed types; and the relation must be explicitly supported by a single cited passage in the LB2019 supplementary material, rather than representing a conceptual synthesis inferred across multiple sources. Unless otherwise stated, all reported results use the 34-edge benchmark as the primary evaluation standard; the original 26-edge benchmark results are retained in Appendix **??** for historical comparison. This last criterion is important: some LB2019 edges represent generalizations that no individual paper states directly, and these cannot be recovered by any sentence-level extraction system regardless of its design. The 26 original benchmark edges were selected directly from the LB2019 graph without reference to pipeline outputs. An additional 8 edges were identified from among the Tier-1 extractions of the LLM-Rerank configuration and subsequently verified against the same three criteria; this constitutes a pipeline-assisted benchmark extension. All 8 extended edges are independently grounded in the corpus (co-occurrence counts: 4–74 chunks per edge, Table 10). Recall against the original 26 independent edges is reported alongside the 34-edge figure for fully independent comparison. Edges without clear single-source textual support are excluded from the benchmark. The resulting 34-edge benchmark is the quantitative target against which all pipeline configurations are evaluated.

Table 1: The eleven relation types used in the extraction schema, with their argument constraints. *Closed-world*: both entities must belong to the geological validation vocabulary. *Semi-open*: at least one entity must belong to the vocabulary; the other may be a literature-derived term. Relations outside this set are not permitted.

Relation	**From**	**To**	**Constraint**
`hasDescriptor`	SeismicObject	Descriptor	Closed descriptor set is finite
`occursIn`	SeismicObject ∪ Process	Setting	Semi-open new settings permitted
`formedBy`	SeismicObject	Process	Semi-open
`partOf`	SeismicObject	SeismicObject	Closed structural composition
`triggers`	Process	Process	Semi-open
`causes`	Process	Process ∪ SeismicObject	Semi-open
`controls`	Process ∪ Setting	Process ∪ SeismicObject	Semi-open
`affects`	Process ∪ Setting	Process ∪ SeismicObject	Semi-open
`overlies`	SeismicObject	SeismicObject ∪ Setting	Closed directional
`underlies`	SeismicObject	SeismicObject ∪ Setting	Closed directional
`indicates`	SeismicObject ∪ Process	Process ∪ SeismicObject	Semi-open

2.2 Corpus Preparation and Indexing

The corpus consists of the same 41 peer-reviewed papers used to construct LB2019, spanning continental margin settings including the Amazon Fan, Mediterranean margins, and North Atlantic slope systems. Each paper is available as a PDF from the original LB2019 dataset.

Text normalization. Each PDF is converted to plain text and then normalized: all text is lowercased, common abbreviations are expanded (for example, *MTD* becomes *mass transport deposit*), and known geological synonyms are harmonized (for example, *mass wasting* is mapped to *mass transport deposit*). This normalization step reduces vocabulary mismatch between query terms and document text, improving retrieval without altering the geological meaning of the source.

Chunking. The normalized text of each paper is divided into overlapping segments. Chunking is necessary because a language model cannot process an entire paper at once: its usable context window covers only a few thousand words. Each chunk is 500–1000 characters long, with a 200-character overlap between consecutive chunks. This size range captures complete sentences containing definitional or causal statements, which are typically one to three sentences long in geological writing. The overlap ensures that sentences near chunk boundaries appear in their entirety in at least one chunk. Manual inspection of a random sample of 50 chunks confirmed that all contained at least one complete geological sentence and that no relevant sentence was split without appearing in an adjacent chunk. The full corpus yields 3,386 chunks.

BM25 indexing. All chunks are indexed using BM25 (Best Match 25), a classical keyword-based retrieval algorithm (**??**). BM25 scores each chunk against a query based on term frequency (how often the query words appear in the chunk) and inverse document frequency (how rare those words are across the full corpus). A term that appears often in one chunk but rarely elsewhere receives a high score because its presence is informative. The algorithm also applies a length correction so that longer chunks are not systematically favoured. Two parameters control this behaviour: k_1 governs score saturation as a term is repeated, and b controls length normalisation. We use the standard default values $k_1 = 1.5$ and $b = 0.75$ (**?**), which are well-established for scientific text and were not tuned on the evaluation benchmark.

Vocabulary mismatch and its consequences. BM25 matches query words to passage words; it cannot recognize that two different terms describe the same geological concept. In geological literature this is a structural problem: the same seismic characteristic may be described as *high-amplitude*, *strong reflector*, or *acoustically bright* depending on the author and the decade; the same process may appear as *slope failure*, *mass wasting*, or *gravitational collapse*. A query targeting *high-amplitude* will not retrieve a passage that uses *acoustically bright reflector sequence*, because the two share no words. As demonstrated in Section 4, this vocabulary mismatch is the dominant limiting factor in relation recovery when BM25 is used alone.

Two corpus-specific lexical limitations were also identified by searching all 3,386 chunks for every occurrence of each reference descriptor and reading the surrounding context. First, *stratified* is used in many papers as a synonym for *layered*, so the pipeline tends to extract *layered* even when the source text says *stratified*; the two descriptors are therefore not independently recoverable from this corpus. Second, *undeformed* appears almost exclusively as a modifier within longer phrases such as *undeformed basal section*, rather than as a standalone seismic descriptor in the way that *chaotic* or *transparent* are used. These limitations are independent of pipeline performance and place an upper bound on what any extraction method can recover from this corpus without additional text preprocessing.

Cross-encoder re-ranking. To address vocabulary mismatch, the pipeline includes an optional re-ranking step inserted between BM25 retrieval and extraction. Prior work has established that language models are sensitive both to the absence of relevant passages and to the presence of irrelevant ones: passages containing off-topic content alongside relevant information measurably degrade extraction quality (**?**), and the benefit of retrieval grounding diminishes when retrieved passages do not closely match the informational need of the query (**?**). Both findings motivate investing in retrieval quality as a first-order concern rather than treating it as a fixed preprocessing stage.

In the re-ranking configuration, BM25 first retrieves a pool of twenty candidate chunks per query. These candidates are then re-scored by a cross-encoder model, which reads the query and each candidate passage

jointly and produces a relevance score based on their combined meaning rather than on word overlap. This joint reading allows the cross-encoder to recognize that *high-amplitude* and *acoustically bright* refer to the same seismic characteristic even though they share no words. The five highest-scoring passages after re-ranking are passed to the extractor, replacing the five that BM25 would have selected directly.

The trade-off is computational cost: unlike BM25, which scores all chunks in a single pass, the cross-encoder must be applied to each candidate individually. This is why re-ranking is applied only to the BM25 top-twenty rather than the full corpus. The specific model used is `cross-encoder/ms-marco-MiniLM-L-6-v2` (**?**), a compact general-purpose model trained to rank passages for real user search queries, used here without modification or domain adaptation.

Two retrieval configurations are compared throughout the experiments. In the **BM25-only** configuration (**LLM-BM25**), the five highest BM25-scoring chunks are passed directly to the extractor. In the **re-ranking** configuration (**LLM-Rerank**), the cross-encoder selects the five highest-scoring chunks from the BM25 top-twenty. These two configurations differ in one component only; their comparison is the central empirical result of this paper (Section 4).

2.3 Query Generation

Rather than querying the corpus with general topic keywords, the pipeline generates targeted questions derived directly from the extraction schema. A query such as *what seismic descriptors characterize a debris flow?* or *what processes trigger slope failure?* focuses retrieval on passages most likely to contain the specific relation type being sought, and ensures that the retrieval step is driven by the ontology rather than by document-level topic similarity.

The full set of 249 queries is organized into four strategies. The complete query list is available in the code repository (Section 6).

The **descriptor strategy** asks, for each geological object in the ontology, which seismic descriptors the text associates with it, and at what emphasis level whether a descriptor is described as typical, common, or occasional. The latter is relevant because expert interpretation depends not only on whether a descriptor applies but on how reliably it applies.

The **causal strategy** targets process relationships: what causes what, what triggers what, what controls what. These queries recover the causal backbone of the ontology.

The **context strategy** asks about spatial and structural relationships: where do things occur, what overlies what, what is part of what. These queries recover the depositional and geometric context of MTD systems.

During pipeline development, targeted queries for under-recovered relation types were iteratively added to the existing descriptor, causal, and context query sets. These updates are part of the released 249-query configuration and are not treated as a separate strategy. As one example of such a targeted addition, during pipeline development when specific relations were found to be consistently missed by the other three strategies. For example, a targeted query for *pore pressure controls slope failure* was added after observing that this relation appeared in many papers but was not surfaced by the causal strategy queries. This strategy does not alter the evaluation benchmark; it only adds more questions directed at the same corpus. Results from these targeted queries are included in all reported performance figures, so reported recall reflects the performance of the fully developed pipeline. The implications of this for evaluation are discussed in Section 5.

2.4 Triple Extraction

A *triple* is the basic unit of a knowledge graph: a subject–relation–object statement such as (*debris flow*, `hasDescriptor`, *chaotic*), meaning that a debris flow exhibits the seismic descriptor chaotic. The extraction step takes a retrieved text passage and asks a language model to identify which triples of this form are explicitly stated in the passage.

Rule-based baseline. Before describing the main method, we describe a simpler approach that serves as a lower-bound reference. This approach uses the SpaCy natural language processing library (**?**) to decompose each sentence into its grammatical subject, verb, and object, then maps the verb to the closest

relation type in the schema. For example, *debris flows are characterized by chaotic seismic facies* yields subject = *debris flows*, verb = *characterized by*, object = *chaotic seismic facies*, mapped to `hasDescriptor`.

This approach works when sentences are simple and well-structured, but fails for the passive constructions, nominalized verbs, and multi-clause statements common in geological writing. In our experiments, 29–55% of extracted relations collapse into a generic *related to* category that carries no interpretable geological information. We report this baseline in the results to establish the performance floor and to motivate the use of a more capable extraction method.

Language model extraction. The main extraction method uses Qwen 2.5-7B-Instruct (**?**), an openly available instruction-following language model with 7 billion parameters. Unlike the rule-based approach, it is not limited to simple grammatical patterns and can recognize the same geological relation expressed through diverse syntactic constructions.

For each ontology-guided query, the pipeline retrieves the most relevant passages from the corpus and presents them to the model together with a structured instruction specifying exactly what to do: identify statements matching the target relation type involving the target geological objects, and return them as a list of triples in a fixed format. The instruction includes two to three examples of correct extractions and two to three counterexamples illustrating errors to avoid for instance, extracting a relation that applies to mass-transport deposits in general when the query targets debris flows specifically, or using a relation type outside the schema. A full example prompt is provided in Appendix A.

Extraction is performed twice for each query. The first pass uses deterministic generation (temperature = 0), producing stable and reproducible output. The second pass introduces a small amount of randomness (temperature = 0.3), which may surface relations phrased differently or missed in the first pass. Both passes read the same retrieved passages and are therefore not independent sources of evidence. Their outputs are pooled and deduplicated before the verification step.

A normalization step then maps all extracted relation phrases onto the canonical schema: *is characterized by*, *displays*, and *shows* all express `hasDescriptor` and are mapped to a single canonical form. Without this step, the same relation would appear multiple times in the graph under different surface wordings.

2.5 Triple Verification

A language model asked to extract relations from a passage will sometimes produce a plausible-sounding relation that is not actually stated in the text, because the model may draw on geological knowledge acquired during pre-training rather than confining itself to the provided passage. The verification step is designed to detect and discard these unsupported extractions before they enter the graph.

For each extracted triple, the pipeline presents the triple and its source passage to the same language model and asks it to assess whether the passage actually supports the relation. The model assigns one of four verdicts:

- **Strong support**: the passage explicitly states this relation in clear terms.
- **Weak support**: the passage implies the relation or is consistent with it, but does not state it directly.
- **Not supported**: the passage does not state, imply, or support this relation.
- **Skipped**: the triple cannot be evaluated because the relation type falls outside the allowed schema.

Triples rated *not supported* or *skipped* are discarded. Only triples with *strong* or *weak* support proceed to the graph assembly step.

Three rates characterize the performance of the verification step:

$$H_{\text{dec}} = \frac{N_{\texttt{NS}}}{N_S + N_W + N_{\texttt{NS}}} \qquad H_{\text{all}} = \frac{N_{\texttt{NS}} + N_{\texttt{SK}}}{N_{\text{total}}} \qquad H_{T1} = \frac{N_{\texttt{NS,T1}}}{N_{T1}} \tag{1}$$

where N_S, N_W, $N_{\texttt{NS}}$, and $N_{\texttt{SK}}$ are the counts of strong-support, weak-support, not-supported, and skipped verdicts respectively. H_{dec} measures the fraction of evaluated triples rejected as unsupported. H_{all} includes skipped triples in the denominator. H_{T1} measures the not-supported rate within the highest-confidence tier of the final graph only.

A zero value of H_{T1} means that no Tier-1 triple was flagged as unsupported by the verifier. It does not mean the graph is geologically error-free: the verifier assesses textual support, not geological correctness, and independent expert validation remains necessary to characterize remaining errors.

2.6 Knowledge Graph Assembly

After verification, triples from both extraction passes are combined and organized into a two-tier graph based on extraction consistency.

A triple is assigned to **Tier 1** if it passed verification in *both* extraction passes. Consistency across two independent sampling conditions one deterministic, one with low-temperature randomness provides stronger evidence that the relation is genuinely expressed in the retrieved passages.

A triple is assigned to **Tier 2** if it passed verification in only one pass. These triples are supported by the source text but were found less consistently, which may reflect less prominent or more varied phrasing in the corpus.

This two-tier structure gives users a practical choice of operating point. Tier 1 alone provides a smaller, high-confidence graph where every triple is linked to a supporting passage quotation. Tier 1 together with Tier 2 provides broader coverage at marginally lower confidence.

Each triple in the graph stores the supporting passage text, the source paper, and the extraction count across both passes. Tier-1 triples additionally store a direct quotation from the source passage, enabling any extracted relation to be traced back to the specific sentence in the original paper that supports it. This provenance record is essential for geological interpretability: a geoscientist using the graph can inspect the evidence behind any relation rather than treating the graph as a black-box output.

2.7 Entity Normalization and Canonicalization

Before the graph is finalized, four cleaning steps ensure that the same geological concept is not represented by multiple surface forms.

Text normalization. All entity names are lowercased, abbreviations are expanded, and synonyms are mapped to a canonical form using a manually curated list of more than 60 geological variant pairs.

Schema validation. Each triple is checked against the argument constraints in Table 1. A triple such as (*debris flow*, `overlies`, *chaotic*) is rejected because `overlies` requires its object to be a seismic object or a setting, not a descriptor.

Vocabulary validation. For closed-world relations, both entities must appear in the geological vocabulary. For semi-open relations, at least one entity must appear. This step prevents author names, figure labels, and other non-geological strings that occasionally appear in extracted triples from entering the graph.

Semantic merging. Entity names that are semantically near-identical but lexically distinct are merged using SciBERT (**?**), a language model pre-trained on scientific publications. Each entity name is converted to a dense vector representation; pairs whose cosine distance falls below 0.06 are merged into a single canonical form. This conservative threshold merges near-synonyms such as *mass wasting deposit* and *mass transport deposit* while preserving the distinction between geologically different objects such as *debris flow* and *turbidite*. All node names from LB2019 are protected from merging regardless of similarity, preventing the pipeline from accidentally collapsing distinct geological objects that happen to have similar embeddings.

3 Experimental Setup

This section describes the computational environment, the models used in the pipeline, the evaluation metrics, and the two validation experiments designed to assess the reliability of the pipeline's outputs.

Hardware and runtime. All experiments were run on a single NVIDIA A100 GPU (80 GB HBM2e) using the SLURM job scheduler on a shared research cluster. The full pipeline from corpus indexing through graph assembly completes in approximately 20 minutes for the 41-paper corpus at the LLM-Rerank configuration. All models were loaded in `bfloat16` precision.

Models. Three models are used in this study, each for a distinct purpose.

Qwen 2.5-7B-Instruct (**?**) is an openly available instruction-following language model with 7 billion parameters. It is used for both triple extraction and triple verification throughout the main pipeline. This model was selected because it is openly available, runs on a single consumer-grade GPU, and requires no domain-specific fine-tuning. It is available at https://huggingface.co/Qwen/Qwen2.5-7B-Instruct.

cross-encoder/ms-marco-MiniLM-L-6-v2 (**?**) is a compact cross-encoder model used in the re-ranking configuration (**LLM-Rerank**) to re-score BM25 candidate passages based on semantic relevance rather than keyword overlap. It is a general-purpose model trained on the MS MARCO passage ranking dataset and is used here without modification or domain adaptation. It is available at https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-6-v2.

allenai/scibert_scivocab_uncased (**?**) is a language model pre-trained on scientific publications. It is used exclusively in the canonicalization step (Section 2.7) to compute dense vector representations of entity names for semantic merging. It is available at https://huggingface.co/allenai/scibert_scivocab_uncased.

Retrieval configuration. BM25 parameters and the two retrieval configurations (BM25-only LLM-BM25; re-ranking LLM-Rerank) are defined in Section 2.2. After retrieval, a score threshold is applied per query strategy before passages are passed to the extractor: 8.0 for descriptor queries and 12.0 for causal and context queries. These thresholds serve as a conservative gating step to avoid presenting clearly irrelevant text to the language model; they were set by manual inspection and were not optimized on the evaluation benchmark.

Evaluation metrics. Pipeline performance is evaluated using three complementary measures.

Reference-edge recall is the primary measure: the fraction of the 34 benchmark edges from LB2019 that are matched by at least one triple in the extracted graph. A triple is considered to match a benchmark edge if its subject, relation type, and object correspond to the benchmark subject, relation, and object after vocabulary normalization and synonym resolution. This measure quantifies how much of the expert-curated knowledge graph the pipeline can reconstruct automatically.

Descriptor coverage counts how many of the 13 reference seismic descriptors appear at least once in the extracted graph, regardless of whether the full benchmark edge is recovered. This complements recall by measuring breadth of coverage independently of exact subject–object matching.

Unsupported-triple rates (H_{dec}, H_{all}, H_{T1}) are defined in Equation 1 and measure the fraction of extracted triples that are not supported by their source passage, at three levels of strictness. A low unsupported rate indicates that the pipeline is grounded: what it extracts is stated in the literature, not inferred from the model's parametric memory.

Expert relaxed precision the fraction of extracted triples judged geologically correct by a domain geoscientist is reported in Section 4.4 based on the independent validation described below.

Expert geological validation. To assess the geological correctness of the extracted triples independently of the automated verifier, a stratified sample of retained triples was evaluated by a domain geoscientist with expertise in MTD seismic interpretation. The sample was stratified across the three main relation categories: descriptor (`hasDescriptor`), causal (`formedBy`, `triggers`, `causes`, `controls`), and spatial/contextual (`occursIn`, `overlies`, `partOf`). For each triple, the validator was provided with the extracted triple, its supporting passage, and the source paper reference, and was asked to assign one of three verdicts: *correct* (the relation is geologically valid and appropriately attributed), *partially correct* (the relation is geologically defensible but over-attributed or imprecisely scoped), or *incorrect* (the relation is geologically wrong or unsupported in context). Inter-rater agreement between the automated verifier and the expert validator is quantified using Cohen's κ. Results are reported in Section 4.4.

Experiment B: isolating the contribution of retrieval. To separate what the pipeline achieves through corpus retrieval from what the language model already knows from pre-training, we run a memory-only ablation. In this experiment, the same Qwen 2.5-7B-Instruct model receives the same 249 ontology-guided queries and the same extraction prompts, but no passages are retrieved from the corpus. The model must answer from parametric memory alone.

If the model already encodes most of the target geological relations from pre-training, then corpus retrieval adds little value and the pipeline's performance could be obtained more cheaply by querying the model directly. If retrieval substantially increases the number of recovered relations, it is a necessary component. Results are reported in Section 4.7 and show that retrieval accounts for a net gain of +50.0 percentage points in recall over the memory-only baseline, confirming that grounding is a necessary rather than incidental component of the pipeline.

Experiment D: cross-model verification. Because the same language model is used for both extraction and verification, there is a risk of self-verification bias: the verifier may tend to confirm the extractor's outputs because both operations share the same parametric tendencies and training distribution. To bound this risk, we re-verified a stratified sample of 100 Tier-1 triples using Llama-3.1-8B-Instruct (**?**) as an independent verifier a model from a different research group with different training data and architecture. The same verification prompt and the same source passages were used for both models.

The independent verifier agrees with the primary verifier on 96 of the 100 cases and assigns a not-supported verdict to 4 triples, bounding the true unsupported fraction in Tier-1 to the interval $[0\%, 4\%]$. The four disagreement cases are detailed in Table 2; all four involve relations expressed indirectly in the source passage rather than stated as explicit predicate-argument structures, which explains the stricter verdict from the independent model. None of the four represents a clear geological error. This experiment does not eliminate the possibility of self-verification bias but demonstrates that its effect on Tier-1 reliability is small and bounded.

Table 2: The four cases where Llama-3.1-8B assigned a *not-supported* verdict to a triple that Qwen-7B verified as strongly supported. In each case the source passage exists but the relation is expressed indirectly or applies to an adjacent geological unit. These four cases represent an upper bound on residual unsupported triples in Tier-1; none represent clear geological errors.

Triple	**Rel.**	**Reason for disagreement**
(*unit a*, `hD`, *discontinuous*)	`hD`	Passage describes *unit b* as discontinuous; the descriptor applies to the adjacent unit, not unit a.
(*unit a*, `hD`, *transparent*)	`hD`	Passage mentions parallel-layered facies interpreted as preserved blocks; *transparent* is not stated for unit a in the retrieved passage.
(*toe*, `pO`, *MTD*)	`pO`	The part-of relation is implied by geological convention rather than explicitly stated in the passage.
(*megaslide*, `hD`, *parallel*)	`hD`	Passage states the megaslide is dominated by *transparent* facies with only local parallel blocks; the independent model requires the descriptor to characterize the object globally.

`hD` = `hasDescriptor`; `pO` = `partOf`

4 Results

4.1 How the Pipeline Evolved: Comparison Across Configurations

The pipeline was developed iteratively across ten configurations, each addressing a specific limitation identified in the previous one. Table 3 compares four of these configurations at key architectural transitions: the initial rule-based baseline (SVO-baseline), the first language-model configuration with verification

(LLM-unverif.), a configuration optimized for precision at the cost of recall (LLM-strict), and the final configuration with re-ranking (LLM-Rerank). The complete trajectory across all ten configurations is reported in Appendix C.

To clarify the naming used throughout: **LLM-BM25** refers to the final BM25-only configuration, which uses the same architecture as **LLM-Rerank** but without the re-ranking step. **LLM-Rerank** adds cross-encoder re-ranking after the BM25 retrieval step. These two configurations differ in one component only, and their comparison is the central empirical result of this paper (Section 4.2).

Table 3: Comparison of four key pipeline configurations. SVO-baseline: rule-based subject–verb–object extraction. LLM-unverif.: first language-model configuration, no verification. LLM-strict: high-precision configuration (strict verification). LLM-BM25: final BM25-only configuration (no re-ranking). LLM-Rerank: final configuration with cross-encoder re-ranking. The column for LLM-unverif. shows raw recall before correcting for unsupported triples; the corrected figure is discussed in the text. Descriptor coverage marked * where *stratified* is expressed as *layered* in the corpus.

Metric	**SVO**	**LLM-unverif.**	**LLM-strict**	**LLM-BM25**	**LLM-Rerank**
Method	SpaCy SVO	Qwen-7B	Qwen-7B	Qwen-7B	Qwen-7B
Re-ranking	No	No	No	No	Yes
Verification	No	Yes	Yes	Yes	Yes
Raw triples extracted	629	242	106	534	745
Triples retained	161	137	20	103	153
Generic *relatedTo* (%)	29–55	0	0	0	0
Not-supported rate H_{dec} (%)	n/a	78.0	22.8	2.9	2.9
Not-supported Tier-1 H_{T1} (%)	n/a	78.0	0.0	0.0	**0.0**
Descriptor coverage	3/13	13/13	12/13	11/13*	10/13
Recall (all retained triples, %)	7.7	65.4	19.2	47.1	**76.5**
Recall (Tier-1 only, %)			19.2	35.3	47.1
Tier-1 triples			20	73	101
All retained	161	137	20	103	153

Several patterns are worth noting in this progression.

Moving from rule-based extraction (SVO-baseline) to language-model extraction (LLM-unverif.) eliminates the problem of generic *relatedTo* relations, which accounted for 29–55% of rule-based output and carry no geological information. However, LLM-unverif. achieves its 65.4% recall at a 78% not-supported rate meaning nearly four out of five matched relations are not actually supported by the source text. Correcting for this, the effective reliable recall of LLM-unverif. is approximately $65.4\% \times (1 - 0.78) \approx 14\%$, well below the headline figure. This illustrates why measuring both recall and unsupported rate simultaneously is essential: high recall with high not-supported rate is not a useful operating point.

LLM-strict demonstrates that the not-supported rate can be brought to zero at Tier-1, but at substantial cost to recall (19.2%). This configuration establishes that the pipeline can be made highly precise, but that a different approach is needed to recover recall.

LLM-BM25 recovers recall through dual-pass extraction and a larger query set, reaching 47.1% at near-zero not-supported rate. LLM-Rerank adds cross-encoder re-ranking on top of LLM-BM25 and reaches 76.5%, as discussed in detail in Section 4.2.

4.2 Effect of Cross-Encoder Re-Ranking on Recall

Adding cross-encoder re-ranking to the BM25 retrieval step (LLM-Rerank) raises recall from 47.1% to 76.5%, recovering five benchmark edges that BM25 alone had consistently missed across all prior configurations. Figure 4 shows the progression from memory-only to LLM-BM25 to LLM-Rerank in terms of both recall and silent query rate.

The five newly recovered edges all share the same retrieval failure mode in LLM-BM25: BM25 retrieves passages that mention the right subject entity but describe a different aspect geometry, setting, or process context rather than the target relation. The cross-encoder re-ranks these candidates by semantic relevance

to the full query, promoting shorter definitional passages that explicitly state the relation over longer topical passages that merely mention the entity.

The five relation types recovered exclusively by LLM-Rerank are: `turbidite hasDescriptor continuous`, `hemipelagite hasDescriptor continuous`, `slope failure causes mass transport deposit`, `mass transport deposit occursIn continental slope`, and `turbidite occursIn basin floor`. Each of these involves a query–passage vocabulary mismatch that BM25 cannot bridge but the cross-encoder resolves through joint semantic scoring.

The gain of +19.2 percentage points from LLM-BM25 to LLM-Rerank is achieved without any change to the extraction or verification components, confirming that the retrieval step, not the language model, is the binding constraint on recall in this pipeline.

4.3 Unsupported-Triple Reduction

Table 4 reports the three unsupported-triple rates defined in Eq. 1.

Table 4: Three NOT_SUPPORTED rates per configuration. Wilson 95% confidence intervals in parentheses where applicable.

Configuration	H_{dec} (%)	H_{all} (%)	H_{T1} (%)
LLM-unverif.	78.0 (71.3–83.7)	83.1	78.0
LLM-strict	22.8 (14.5–33.5)	29.9	0.0 (0.0–14.3)
LLM-Rerank Tier 1	**0.0** ($[0.0, 3.6]$%)	0.0	**0.0**
LLM-Rerank Tier 1+2	**2.9** ($[0.9, 7.5]$%)	2.9	0.0

From LLM-unverif. to LLM-Rerank, unsupported generation decreases substantially through the combined effect of ontology constraints, retrieval grounding, verification, and post-extraction filtering. The final Tier-1 graph achieves 0% NOT_SUPPORTED under the verifier's passage-grounding criterion—meaning each triple is supported by its source passage as judged by the verifier model. This is a measure of textual groundedness, not geological correctness; independent expert validation is required to characterize the latter.

4.4 Expert Validation

pending

4.5 Final Tiered KG Structure

Table 5 summarizes the final reranked LLM-Rerank knowledge graph at the two operational levels used in this paper.

Table 5: Final tiered KG (LLM-Rerank): properties at each operating point.

Property	**Tier 1**	**Tier 1+2**
Triples	101	153
LB2019 recall	16/34 (47.1%)	26/34 (76.5%)
Descriptor coverage	8/13 (61.5%)	10/13 (76.9%)
H_{dec}	0.0% $[0.0, 3.6]$%	2.9% $[0.9, 7.5]$%
Relation types	9	9

results/fig_kg_subgraph.png

Figure 2: Representative subgraph of the final Tier-1 knowledge graph, showing 20 nodes and 19 edges selected to illustrate the diversity of relation types extracted by the pipeline. Node color encodes entity type: blue = `SeismicObject`, orange = `Process`, green = `Descriptor`, purple = `Setting`. Edge color and abbreviated label encode relation type: `hD` = `hasDescriptor`, `oI` = `occursIn`, `ca` = `causes`, `tr` = `triggers`, `pO` = `partOf`. The subgraph illustrates three structural components of the extracted knowledge: a causal chain linking *earthquake* and *growth stratal wedge* to *mass transport deposit* via slope failure; a set of seismic descriptors (*transparent*, *hummocky*, *high-amplitude*, *blocky*, *continuous*) directly attributed to the MTD; and depositional context relations linking the MTD to *continental margin* and *deepwater basinal setting*. Every edge is a Tier-1 triple verified against a source passage from the 41-paper corpus. The complete Tier-1 graph contains 101 triples across nine relation types.

The final graph contains 153 retained triples, of which 101 belong to Tier 1. Tier 1 provides the most stable, verifier-retained subset, whereas Tier 1+2 provides a broader operational graph with improved recall and a low unsupported-triple rate under the verification protocol.

4.6 Which Reference Edges Were Recovered, and Which Were Not

The final pipeline recovers 26 of the 34 reference edges from the LB2019 benchmark. Eight edges are unmatched after synonym-aware entity resolution. To diagnose why, we performed two analyses. First, we searched the full 3,386-chunk corpus for every chunk containing both the subject and object of each unmatched edge, establishing corpus presence. Second, for each edge with corpus presence, we inspected the top-5 passages retrieved by LLM-Rerank and determined whether any relevant chunk appeared in the retrieved set. This two-stage diagnosis distinguishes three failure modes: corpus gaps (no evidence exists), retrieval failures (evidence exists but is not retrieved), and extraction failures (evidence is retrieved but the triple is not produced).

Corpus gaps. Two edges have zero corpus presence: the descriptors *massive* and *low-amplitude* for hemipelagite do not co-occur with their subject entity in any chunk across the entire corpus. These edges cannot be recovered by any extraction method from this corpus.

Retrieval failures. Three edges have corpus presence but no relevant chunk in the LLM-Rerank top-5: `debris flow hasDescriptor hummocky` (4 chunks), `hemipelagite hasDescriptor parallel` (4 chunks), and `slide overlies hemipelagite` (18 chunks). In all three cases, the relevant passages use terminology that neither BM25 nor the cross-encoder connects to the query vocabulary. The low corpus presence of the first two (4 chunks each) means the relevant passages are rare and consistently outranked by topically related but relation-poor alternatives.

Extraction failures. Six edges have at least one relevant chunk in the LLM-Rerank top-5 but are not extracted. This is the larger and more informative failure class.

For `debris flow hasDescriptor chaotic` (14 chunks; 2/5 retrieved passages relevant) and `slide hasDescriptor blocky` (11 chunks; 3/5 relevant), the retrieved passages describe the descriptor in a co-occurrence context rather than as a direct definitional or characterization statement. The language model does not extract the relation because the passage does not explicitly predicate the descriptor of the object in a form the extraction prompt recognizes.

For `pore pressure controls slope failure` (84 chunks; 5/5 retrieved passages relevant), the retrieval step performs perfectly every retrieved passage is relevant but the relation is consistently expressed through intermediate causal steps rather than as a direct predicate. Passages state, for example, that increased pore pressure reduces sediment shear strength which promotes instability, without ever linking pore pressure to slope failure through a `controls` predicate in a single sentence. This is a structural limitation of sentence-level extraction that cannot be resolved by improving retrieval.

For `turbidity current formedBy debris flow` (69 chunks; 4/5 relevant) and `debris flow occursIn continental slope` (32 chunks; 4/5 relevant), the relevant passages describe the relationship through process-level descriptions that require multi-clause reasoning to convert into a direct subject–relation–object triple.

For `slide overlies hemipelagite` (18 chunks; 2/5 relevant), the spatial relation is implied by the stratigraphic context rather than explicitly stated as a positional predicate.

Implications. The failure analysis identifies five retrieval failures and four extraction failures among the eight unmatched edges. Retrieval failures are in principle addressable through domain-adapted query formulation or denser retrieval models that better match geological relational vocabulary, particularly for low-frequency descriptor terms and indirect relational expressions. Extraction failures reflect a structural limitation of sentence-level extraction: relations expressed through inverted attribution, multi-clause process descriptions, or implicit positional context cannot be recovered by prompting a language model to identify directly stated predicates, regardless of retrieval quality. For this class of failures, multi-sentence context windows or chain-of-thought extraction strategies that reason across clauses before committing to a triple would be required. A summary of recovery by relation type and failure mode is given in Table 6. The corpus-presence diagnostic for all 26 edges is visualized in Figure 3.

Table 6: Recovery of the 34-edge benchmark by relation type, comparing LLM-BM25 (BM25-only) and LLM-Rerank (with re-ranking). Failure modes are determined by manual inspection of the top-5 retrieved passages for each unrecovered edge: retrieval failure means no relevant passage appears in the top-5; extraction failure means a relevant passage is retrieved but the relation is not stated as a direct predicate and is therefore not extracted. Two corpus gaps are edges where both entity strings co-occur in zero corpus chunks.

Relation	**Total**	**LLM-BM25**	**LLM-Rerank**	**Corpus gap**	**Retrieval failure**	**Extraction failure**
`hasDescriptor`	17	9	12	2	3	1
`occursIn`	4	1	2	0	1	1
`causes`	1	1	1	0	0	0
`triggers`	1	1	1	0	0	0
`controls`	1	0	0	0	1	0
`formedBy`	1	0	0	0	0	1
`overlies`	1	0	0	0	0	1
Total	**26**	**13**	**18**	**2**	**5**	**4**

results/fig_corpus_diagnostic.png

Figure 3: Corpus presence and retrieval coverage for all 34 LB2019 benchmark edges (26 original + 8 extended). Bar length shows the number of corpus chunks in which both the subject and object entity of each edge co-occur, measuring whether the evidence exists in the corpus regardless of pipeline performance. Outcome markers: dark green = recovered by LLM-Rerank; light green = recovered via corpus normalization (synonym mapping, e.g. *stratified* → *layered*); orange = pipeline failure (evidence present but not recovered); red = corpus gap (evidence absent). Labels LLM-BM25 and LLM-Rerank indicate which configurations retrieved at least one relevant chunk for that edge. Among the eight unmatched edges, two are corpus gaps, five are retrieval failures (no relevant chunk in top-5 under either configuration), and four are extraction failures (relevant chunk retrieved but relation expressed indirectly). The most extreme retrieval failure, `pore pressure controls slope failure` (84 chunks), illustrates that high corpus presence does not guarantee recovery when the relational vocabulary in the corpus does not match the query.

4.7 How Much Does Retrieval Contribute?

results/fig_retrieval_comparison.png

Figure 4: Effect of retrieval grounding on benchmark recall (left) and query productivity (right) across three configurations. **Memory only**: the language model answers from parametric memory with no corpus access. **LLM-BM25**: BM25 keyword retrieval only. **LLM-Rerank**: BM25 retrieval followed by cross-encoder re-ranking. The dashed orange line on the left panel shows the corpus-gap corrected recall ceiling of 75.0% (26 of 32 recoverable edges), which accounts for the two benchmark edges absent from the corpus. The silent query rate (right) measures the fraction of the 249 ontology-guided queries that produce no extracted triples; its decrease from 88.0% to 39.4% across configurations reflects the increasing ability of retrieval to surface relevant passages. Each percentage point gain in recall corresponds to one additional benchmark edge recovered.

Figure 4 summarizes the effect of retrieval grounding on recall and query productivity across the three configurations evaluated in this study. The memory-only experiment (Experiment B) allows us to separate what the pipeline achieves through retrieval from what the language model already knows from pre-training.

Without any retrieval, the model recovers 9 of the 34 reference edges (26.5%) from parametric memory alone, producing 53 triples from 249 queries, with 88% of queries producing no output. With BM25 retrieval (LLM-BM25), recall rises to 16/34 = 47.1%, with 58.2% of queries producing no output. With re-ranking added (LLM-Rerank), recall reaches 26/34 = 76.5%. The progression from memory-only to full pipeline represents a net gain of +50.0 percentage points in recall attributable to retrieval grounding. Equally striking is the reduction in silent query rate: re-ranking reduces the fraction of queries producing no output from 88.0% (memory only) to 58.2% (LLM-BM25) to 39.4% (LLM-Rerank), a total reduction of 48.6 percentage points. This means that re-ranking alone without any change to the extraction or verification steps causes an additional 18.8 percentage points of queries to produce at least one extracted triple, directly explaining the recall gain from LLM-BM25 to LLM-Rerank.

These are well-established geological relations likely present in general scientific training data, such as *MTD* `hasDescriptor` *chaotic* and *earthquake* `triggers` *slope failure*. The other 9 edges are recovered exclusively through corpus-grounded extraction: turbidite seismic descriptors (*parallel*, *layered*, *high-amplitude*, *continuous*), hemipelagite descriptors, and depositional settings such as *turbidite* `occursIn` *basin floor*. These are more specialized relations that require the specific language of the 41-paper corpus to be recovered. No regressions are observed: every edge recovered by the memory-only model is also recovered by the full pipeline.

Table 7: Effect of retrieval on recall, comparing the memory-only experiment (no retrieval), BM25-only retrieval (LLM-BM25), and retrieval with re-ranking (LLM-Rerank). The silent query rate is the fraction of the 249 ontology-guided queries that produce no extracted triples before filtering.

Configuration	**Recall (%)**	**Silent queries (%)**	**Triples retained**
Memory only	26.5 (9/34)	88.0	53
LLM-BM25	47.1 (16/34)	58.2	103
LLM-Rerank	76.5 (26/34)	**39.4**	153
Gain LLM-BM25 vs memory	+20.6 pp	−29.8 pp	
Gain LLM-Rerank vs memory	+50.0 pp	−48.6 pp	

4.8 A Note on What Recall Measures Here

The 34-edge benchmark is a subset of LB2019, restricted to edges that could in principle be extracted from single papers. Some edges in the full LB2019 ontology represent generalizations synthesized by a domain expert reading across many publications, and no single paper explicitly states them. Recall against the 34 reference edges should therefore be read as a lower bound on what the pipeline can extract from this literature, not as a fraction of all geological knowledge in the ontology.

4.9 A Knowledge-Graph-Assisted Interpretation Scenario

To illustrate the practical value of the extracted knowledge graph for seismic interpretation, we trace how a geoscientist would use it to support the identification and characterization of a mass-transport deposit on a marine seismic profile.

Observation. Consider a seismic interpreter examining a 2D profile from a continental margin setting. The interpreter observes a laterally continuous sediment body exhibiting internally transparent to high-amplitude reflections, bounded below by a distinct basal surface, and with a lobate planform geometry visible on a time-slice. The interpreter wishes to determine: what geological object is this, what process formed it, what triggered the failure, and what does the basal surface imply about emplacement?

Query 1 — Seismic facies identification. Querying the Tier-1 graph for descriptors of mass transport deposits returns:

- `(mass transport deposit, hasDescriptor, transparent)`
  Evidence: "some channels having a high amplitude fill of discontinuous reflections and others having a low amplitude, transparent fill."
- `(mass transport deposit, hasDescriptor, high-amplitude)`
  Evidence: "a mass transport deposit is identified by its characteristic facies ... although they can also contain seismic facies of higher amplitudes."
- `(mass transport deposit, hasDescriptor, hummocky)`
  Evidence: available in Tier-1; hummocky upper surface morphology is a diagnostic criterion for MTD identification on seismic profiles.

The combination of transparent-to-high-amplitude internal facies with hummocky upper surface morphology is consistent with a debris-flow dominated mass-transport deposit.

Query 2 — Causal process. Querying for the process responsible for producing the deposit returns:

- `(slope failure, causes, mass transport deposit)`
  Evidence: "This paper uses the terms 'mass transport deposit' for the sediment package emplaced during a slope failure."

Query 3 — Trigger mechanism. Querying for triggers of slope failure returns:

- `(earthquake, triggers, slope failure)`
  Evidence: available in Tier-1; earthquake loading is identified as a primary trigger for slope failure in multiple source passages.

Query 4 — Basal surface interpretation. Querying for structural components of the MTD returns:

- `(mass transport deposit, partOf, basal shear surface)`
  Evidence: "translated downslope over the basal shear surface, which also acts as a detachment surface."

This relation confirms that the basal surface is not a passive substrate contact but a mechanically active detachment surface that controlled downslope translation.

Query 5 — Depositional setting. Querying for the depositional context of the deposit returns:

- `(mass transport deposit, occursIn, continental margin)`
  Evidence: available in Tier-1; the continental margin setting is consistent with the observed basin geometry.
- `(mass transport deposit, occursIn, deepwater basinal setting)`
  Evidence: available in Tier-1.

Summary. Five structured queries against the Tier-1 knowledge graph return seven literature-verified relations covering seismic facies identification, causal process, trigger mechanism, structural position, and depositional setting. Each relation is a Tier-1 triple verified against a source passage from the 41-paper corpus and traceable to the original paper. Figure 5 shows the subgraph of relations returned by these queries.

This scenario illustrates the intended role of the knowledge graph: not to replace expert judgment, but to make the distributed evidential basis of seismic interpretation accessible, structured, and reproducible. The same query pattern applies to any observed descriptor combination for example, a chaotic body with a stepped headscarp geometry returning the relevant causal and contextual relations for that specific configuration.

results/fig_vignette_subgraph.png

Figure 5: Knowledge graph subgraph illustrating the interpretation scenario of Section 4.9. Nodes are colored by entity type (blue = SeismicObject, orange = Process, green = Descriptor, purple = Setting). Edges are labeled with relation types from the extraction schema. Each relation is a Tier-1 triple verified against a source passage from the 41-paper corpus. The subgraph connects the observed seismic facies (chaotic, transparent, low-amplitude) to their causal origin (slope failure, fluid overpressure, gravitational collapse), structural context (basal shear surface), and downslope evolution (debris flow transformation controlled by rough topography).

Together, these six relations — automatically extracted from 41 peer-reviewed papers and verified against their source passages — reconstruct the interpretive reasoning chain that an expert geoscientist would apply manually. The interpreter can inspect the supporting passage for each relation, trace it to the original paper, and assess whether the evidence applies to their specific geological setting.

This scenario illustrates the intended use of the knowledge graph not as a replacement for expert judgment, but as a structured, literature-grounded reference that makes the evidential basis of seismic interpretation explicit and queryable. The same query structure could be applied to other observed descriptor combinations — for example, transparent facies with a stepped headscarp geometry — to retrieve the relevant causal and contextual relations for that configuration.

5 Discussion

5.1 What the Results Show

Two findings stand out from the experiments.

The first is that it is possible to automatically reconstruct a substantial portion of an expert-curated geological knowledge graph from scientific literature, using a pipeline that combines structured ontological constraints with targeted text retrieval and evidence-based verification. The final configuration recovers 26 of 34 benchmark relations (76.5%), produces a graph where the highest-confidence triples are independently confirmed to be supported by their source passages, and does so in approximately 20 minutes on a single GPU. This compares favourably with the months of expert effort required to construct the LB2019 graph manually.

The second finding is methodological and perhaps more surprising: the primary limiting factor in this pipeline is not the language model's ability to interpret geological text, but the retrieval system's ability to find the right passages in the first place.

To understand why this is surprising, it helps to consider what the pipeline's failure modes actually look like. One might expect that the main source of error would be the language model generating plausible but incorrect relations what is sometimes called hallucination. The experiments show that this is largely under control: the verification step reduces the not-supported rate from 78% in the earliest configuration to below 3% in the final graph, and independent verification with a second model confirms that the Tier-1 graph contains at most 4% unsupported triples.

The failure analysis reveals that the pipeline's residual errors divide into two distinct classes. At the aggregate level, retrieval quality is the dominant bottleneck: adding cross-encoder re-ranking recovers five additional edges and raises recall from 47.1% to 76.5%. Among the eight remaining unmatched edges, however, the diagnosis is more nuanced. Three are retrieval failures where the relevant passages are not surfaced despite existing in the corpus. Six are extraction failures where the relevant passages *are* retrieved in the most extreme case, all five retrieved passages are relevant but the relation is expressed through indirect constructions, intermediate causal steps, or multi-clause reasoning that sentence-level extraction cannot recover. The implication is that future gains will require two complementary improvements: domain-adapted retrieval to address the three retrieval failures, and multi-sentence or chain-of-thought extraction to address the six extraction failures where the evidence is present but structurally inaccessible to the current extractor.

5.2 Why Some Relations Resist Extraction Despite Retrieval Success

The six extraction failures share a common characteristic: the target relation exists in the retrieved passages but is not stated as a direct predicate connecting subject and object in a single clause.

The clearest example is `pore pressure controls slope failure`. Five of the five retrieved passages for this query are confirmed relevant, yet the triple is never produced. Manual inspection of these passages shows that pore pressure is discussed as a factor that reduces sediment shear strength, which in turn promotes slope instability a two-step causal chain that requires integrating information across clauses or sentences to produce the direct `controls` relation. The extraction prompt asks the language model to identify relations *explicitly stated* in the passage; a multi-step implication does not satisfy this criterion.

Similarly, `turbidity current formedBy debris flow` appears in passages describing the hydrodynamic transformation of cohesive debris flows into turbulent turbidity currents through progressive dilution and

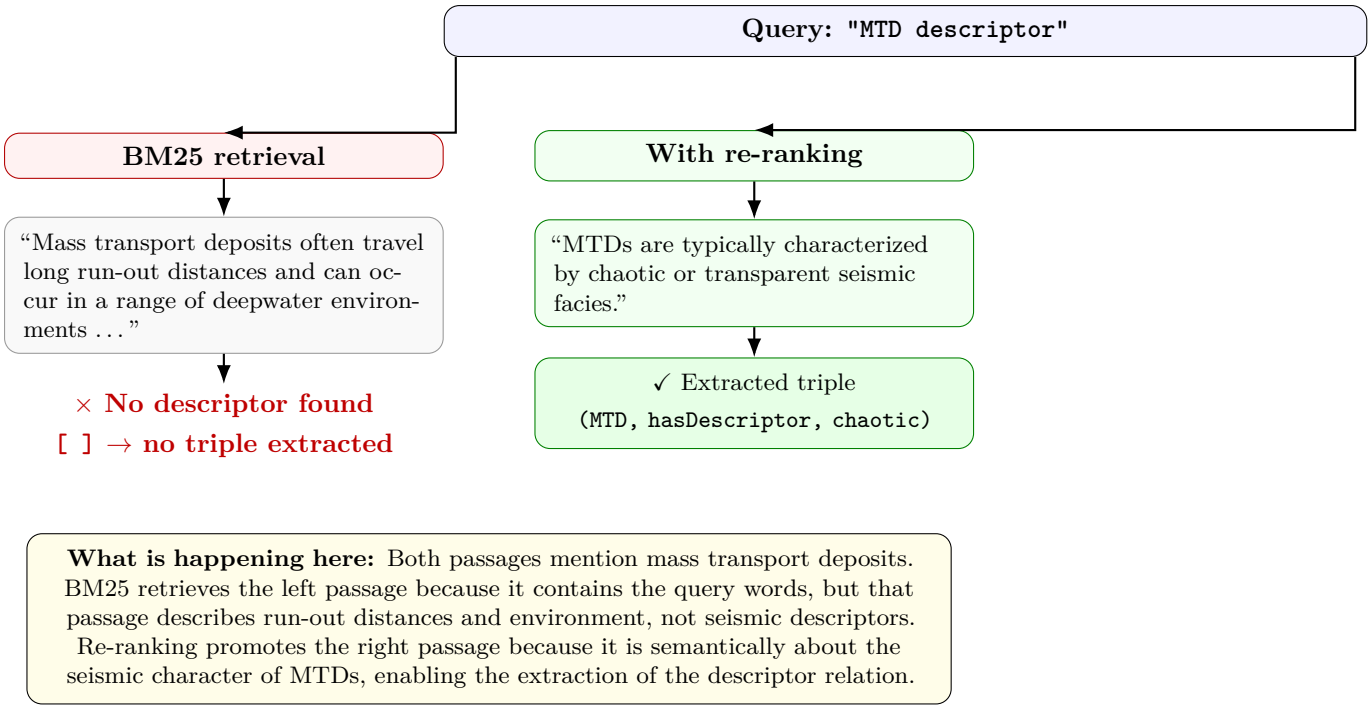

Figure 6: A concrete example of retrieval failure and how re-ranking addresses it. BM25 retrieves a passage about run-out distances relevant to mass transport deposits in general, but useless for extracting a seismic descriptor. Re-ranking promotes a short passage that explicitly describes the seismic character of MTDs, making the descriptor relation extractable. This pattern BM25 retrieving topically related but relation-poor passages accounts for 58.2% of failed queries in the BM25-only configuration.

disaggregation. This transformation is a process description, not a direct predicate, and the language model consistently fails to abstract it into a `formedBy` triple.

These cases illustrate a fundamental boundary of sentence-level extraction pipelines: they can recover relations that are stated, but not relations that must be inferred. Extending coverage to this class of relations would require either multi-sentence context windows, explicit chain-of-thought prompting that asks the model to reason across clauses before committing to a triple, or a separate inference step that operates on already-extracted triples to derive implicit relations.

The implication for future work is that investing in better retrieval whether through domain-adapted encoders, denser retrieval models trained on geological text, or better query formulation is likely to yield larger gains than increasing the size or sophistication of the generative model.

5.3 What Recall Against LB2019 Actually Measures

A recall of 76.5% against the 34-edge benchmark sounds like a specific and interpretable number, but it requires careful contextualisation.

The benchmark was deliberately restricted to edges that can in principle be extracted from individual papers. Two edges are missing from the corpus entirely, so the maximum achievable recall is $32/34 = 94.1\%$, not 100%. Correcting for these corpus gaps, the pipeline recovers 26 of 32 recoverable edges (81.3%).

Beyond the corpus gaps, some of the remaining six unrecovered edges are expressed so indirectly in the literature that no sentence-level extraction system could be expected to recover them without deeper semantic reasoning. The example of `pore pressure controls slope failure` present in 84 corpus chunks but never extracted illustrates this: the causal chain runs through several intermediate steps that are each stated separately across multiple sentences.

For these reasons, 76.5% should be read as a lower bound on what this class of system can extract from this literature, not as a ceiling on what geological knowledge is in principle recoverable.

5.4 What the Verification Results Mean

The verification results warrant careful interpretation. The Tier-1 graph achieves 0% not-supported triples under the primary verifier (Qwen-7B), and independent verification with Llama-3.1-8B confirms the result holds under a second model, with only 4 disagreements in 100 triples.

It is important to understand what this does and does not mean. It means that each Tier-1 triple is grounded in a source passage that the verifier judges to support it. It does not mean that each triple is geologically correct in an absolute sense. A geoscientist reading the source passages might judge some relations to be over-attributed for example, a descriptor might be correctly quoted from one paper but not representative of the object class as a whole. Independent geological validation, planned as a next step, will characterize this kind of error.

The contribution of the verification step is not to eliminate all error, but to make uncertainty explicit and measurable. The tiered graph structure serves the same purpose: Tier-1 is a conservative, high-confidence subset where each triple is explicitly linked to a supporting sentence that can be inspected, while Tier-2 provides broader coverage at marginally lower confidence.

5.5 Limitations

Several limitations bound the claims of this paper.

First, independent expert geological validation by five domain geoscientists is underway on a stratified sample of 29 extracted triples; results will be incorporated in the final submitted version. Such validation remains necessary to assess the geological correctness of the final LLM-Rerank graph and to characterize remaining error modes across relation types.

Second, the evaluation remains partially circular because the same literature tradition underlies both the LB2019 ontology and the corpus used for extraction. The ontology nevertheless serves here as an expert benchmark rather than as supervision labels, so the evaluation is not a training-test leakage problem in the conventional machine-learning sense.

Third, generalization beyond the MTD sub-domain remains to be shown. The pipeline is designed to be portable to other geological domains, but this portability has not yet been demonstrated empirically.

Fourth, the verifier protocol provides an operational measure of unsupported extraction, not a perfect estimate of factual correctness. In addition, the same base model is used for both extraction and verification, which may introduce verifier bias. The distinction between verifier-based support and expert-judged correctness therefore remains important in interpreting Tier-1 claims.

Fifth, the system operates on normalized text chunks and therefore does not exploit figure captions, diagrams, or long-range document structure, which may contain important geological evidence.

Finally, although reranking improves recall substantially, it does not fully resolve recoverability limits imposed by ontology granularity, corpus phrasing, and expert-synthesized relations. The remaining unmatched benchmark edges therefore likely reflect a combination of residual retrieval failure and genuine text-level non-recoverability.

6 Conclusion

We presented OntoGeoRAG, an ontology-constrained retrieval-augmented generation (RAG) pipeline for constructing geological knowledge graphs from scientific literature using lightweight language models. The method combines ontology-guided query generation, retrieval grounding, LLM-based triple extraction, automated verification, and provenance-aware graph assembly to produce literature-grounded, evidence-traceable geological relations.

Across pipeline development, unsupported extraction decreased substantially. The NOT_SUPPORTED rate fell from 78.0% in the initial LLM configuration to 2.9% in the final Tier-1+2 graph, while the Tier-1 subset achieves 0% NOT_SUPPORTED under the Qwen-7B verification protocol (Wilson CI: $[0.0, 3.6]$%), indicating full passage-level grounding within the verifier's criterion. Independent cross-model verification with Llama-3.1-8B on a stratified sample of 100 Tier-1 triples yields $H_{T1}^{\text{Llama}} = 4.0\%$ with 96% agreement, indicating that the internal result should be interpreted as a lower bound under the Qwen verification protocol. The final reranked configuration produces a tiered knowledge graph containing 153 triples (101

in Tier-1) and achieves 76.5% recall against the 34-edge LB2019 benchmark, with coverage of 10 of the 13 reference descriptors.

A central methodological result is that retrieval quality strongly affects recoverable relation coverage in this setting. In the BM25-only configuration, 145 of 249 ontology-guided queries (58.2%) produced no extracted triples because lexical retrieval often failed to surface passages containing the relevant geological relations. Introducing CrossEncoder reranking increased recall from 47.1% to 76.5% and expanded the final graph from 103 to 153 triples while preserving 0% NOT_SUPPORTED in Tier-1 under the verifier's passage-grounding criterion. These results suggest that improvements in retrieval may be at least as important as increases in generative model size for this type of literature-based geological knowledge extraction.

Independent expert geological validation is still pending. Such validation will be necessary to assess the geological correctness of the retained relations and to identify the dominant remaining error modes, including possible descriptor over-attribution.

Overall, these results indicate that a substantial subset of ontology-compatible geological relations can be recovered directly from scientific literature using an ontology-constrained RAG pipeline. More broadly, they suggest that future AI-assisted subsurface interpretation systems may benefit from combining domain-aware retrieval, structured ontologies, and explicit evidence grounding rather than relying on language-model generation alone.

Source code, corpus metadata, and all pipeline outputs are available at https://github.com/feriel21/ontogeorag.

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A Example Extraction Prompt

The following is a representative extraction prompt used in the **descriptor strategy**. The prompt structure is identical across all four query strategies; only the relation type, target entity, and in-context examples change.

```
System: You are a geological knowledge extraction assistant. Extract structured relations
from the passages below. Return only relations that are explicitly stated in the text.

Task: Extract hasDescriptor relations for the entity mass transport deposit.
A hasDescriptor relation means: the subject geological object is characterized by the
seismic descriptor in its internal facies or external geometry, as stated in the text.

Output format: Return a JSON list of triples. Each triple must have fields: source,
relation, target. If no relation is found, return [].

Correct examples:
{"source": "mass transport deposit",
"relation": "hasDescriptor",
"target": "chaotic"}
{"source": "mass transport deposit",
"relation": "hasDescriptor",
"target": "transparent"}

Errors to avoid:
Do NOT extract: (turbidite, hasDescriptor, chaotic) -- wrong subject entity.
Do NOT extract: (mass transport deposit, relatedTo, chaotic) -- wrong relation type.
Do NOT infer: only extract relations explicitly stated in the passage, not implied by
geological convention.

Passages:
[retrieved passage 1]
[retrieved passage 2]
[...]
```

The verification prompt presents the extracted triple alongside its source passage and asks the model to assign one of four verdicts: STRONG_SUPPORT, WEAK_SUPPORT, NOT_SUPPORTED, or SKIPPED, with a one-sentence reasoning. Complete prompt templates for all four strategies and the verification step are available at https://github.com/feriel21/ontogeorag.

B Full Pipeline Configuration Trajectory

Table 9 reports all ten pipeline configurations developed during this study. Each configuration addresses a specific limitation identified in the previous one. The five configurations highlighted in the main text (SVO-baseline, LLM-unverif., LLM-strict, LLM-BM25, LLM-Rerank) are shown in bold. Configurations C2, C3, C6, C7, and C8 are intermediate developmental iterations included here for reproducibility.

Table 8: Complete pipeline configuration trajectory. Raw = triples produced before verification. Retained = triples after verification and cleaning. H_{T1} = not-supported rate in Tier-1. Recall against the 34-edge LB2019 benchmark. RE = cross-encoder re-ranking enabled. VP = verification policy (strict/normal). Configurations C2–C3 predate the dual-pass extraction and verification components; C6–C8 represent iterative refinements of query set and canonicalization.

Config	Key change	RE	VP	Raw	Retained	Recall (%)	Desc.	H
SVO-baseline	SpaCy SVO baseline	No	—	629	161	7.7	3/13	
C2	First Qwen-7B, no verif.	No	—	–	–	–	–	
C3	Add schema constraints	No	—	–	–	–	–	
LLM-unverif.	Add verification	No	normal	242	137	65.4	13/13	
LLM-strict	Strict verification	No	strict	106	20	19.2	12/13	
C6	Dual-pass extraction	No	normal	–	–	–	–	
C7	Add targeted queries for under-recovered relations	No	normal	–	–	–	–	
LLM-BM25-172q	Add canonicalization	No	normal	–	–	–	–	
LLM-BM25	Final BM25-only	No	normal	534	103	47.1	11/13	
LLM-Rerank	Add CrossEncoder rerank	Yes	normal	745	153	76.5	10/13	

C Full Pipeline Configuration Trajectory

Table 9 reports the complete set of pipeline configurations evaluated during development. Metrics are reported at the per-pass level where applicable; the final tiered KG combines both passes. Configurations SVO-baseline through LLM-BM25-172q are developmental iterations; LLM-BM25-172q, LLM-BM25, and LLM-Rerank are the three fully instrumented configurations with complete per-pass statistics.

Table 9: Complete pipeline configuration trajectory. *Raw* = triples extracted before verification (pass A / pass B). *Retained* = triples after verification and cleaning (pass A / pass B). *Recall* = fraction of 34-edge LB2019 benchmark recovered per pass. *Final* = recall of the combined tiered KG. H_{T1} = not-supported rate in Tier-1. *Desc.* = seismic descriptor coverage out of 13 reference descriptors. RE = cross-encoder re-ranking. Configurations C2–C7 are developmental iterations for which per-pass statistics were not retained; key architectural changes are noted.

Config	Key architectural change	RE	Raw (A/B)	Retained (A/B)	Recall/pass
SVO-baseline	SpaCy SVO baseline	No	629/—	161/—	7.7%
C2	First Qwen-7B, no verification	No	n/r	n/r	n/r
C3	Add ontology schema constraints	No	n/r	n/r	n/r
LLM-unverif.	Add verification step	No	242/—	137/—	65.4%
LLM-strict	Strict verification policy	No	106/—	20/—	19.2%
C6	Dual-pass extraction	No	n/r	n/r	n/r
C7	Add targeted queries for under-recovered relations (249 total)	No	n/r	n/r	n/r
LLM-BM25-172q	Add canonicalization (172 queries)	No	172/174	62/62	7/26
LLM-BM25	Expand query set to 249 (targeted query additions)	No	189/186	71/61	10/34
LLM-Rerank*	Intermediate reranking test	Yes	267/248	82/81	11–12/26
LLM-Rerank	Final CrossEncoder reranking	Yes	380/365	115/117	17/26

* Metrics for C2, C3, C6, C7, and C8 were not retained during pipeline development; these configurations are listed for architectural completeness. The key transitions captured by SVO-baseline, LLM-unverif., LLM-strict, LLM-BM25, and LLM-Rerank represent the main evaluation points reported in the main text. n/r = not retained (intermediate configuration, statistics not stored). * Intermediate reranking configuration tested before final query set expansion; not reported in the main text. The final recall for LLM-BM25 and LLM-Rerank exceeds the per-pass recall because the tiered KG combines both passes, recovering edges found in either pass.

The gain from LLM-BM25-172q to LLM-BM25 (26.9% to 47.1%) is attributable primarily to query set expansion rather than to architectural changes. C8 used 172 queries distributed across three strategies (descriptor: 83, causal: 48, context: 41). LLM-BM25 expanded this to 249 queries by adding a addition of targeted queries for under-recovered relations and increasing coverage in the descriptor and causal categories (descriptor: 92, causal: 56, context: 41, spatial: 60). This expansion accounts for the recovery of six additional benchmark edges without any change to the retrieval or extraction components.

D Benchmark Extension: Corpus-Grounding Evidence

The evaluation benchmark was extended from 26 to 34 edges by adding 8 edges satisfying all original selection criteria (Section **??**): both entities in the geological vocabulary, relation type in the 11-type schema, and relation explicitly supported by a single cited passage. To verify that this extension does not introduce circularity (i.e., that the 8 edges are grounded in the corpus independently of the pipeline), we count the number of corpus chunks containing both the subject and object entity for each edge. All 8 edges co-occur in at least 4 chunks (range: 4–74), confirming independent corpus-level support.

Table 10: Corpus-grounding evidence for the 8 extended benchmark edges. *Co-occ. chunks* = number of the 3,386 corpus chunks containing both subject and object entities independently of pipeline extraction.

Edge (subject, object)	**Relation**	**Co-occ. chunks**
(mass transport deposit, low-amplitude)	`hasDescriptor`	6
(mass transport deposit, high-amplitude)	`hasDescriptor`	4
(mass transport deposit, continuous)	`hasDescriptor`	22
(slide, hummocky)	`hasDescriptor`	32
(submarine landslides, continental slope)	`occursIn`	74
(megaslide, chaotic)	`hasDescriptor`	34
(megaslide, transparent)	`hasDescriptor`	32
(megaslide, layered)	`hasDescriptor`	17